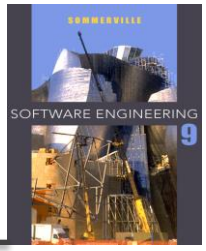


Lecture 2 – Software Processes

Part 1

Source : https://ifs.host.cs.st-andrews.ac.uk/Books/SE9/Presentations/index.html

Topics covered

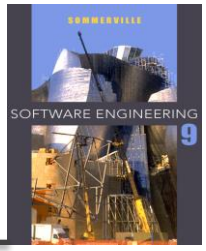


- ✧ Software process models
- ✧ Process activities
- ✧ Coping with change
- ✧ The Rational Unified Process

What is Software Engineering?

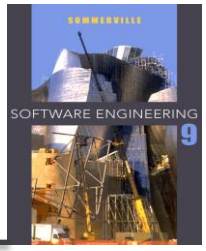


What is Software Engineering?



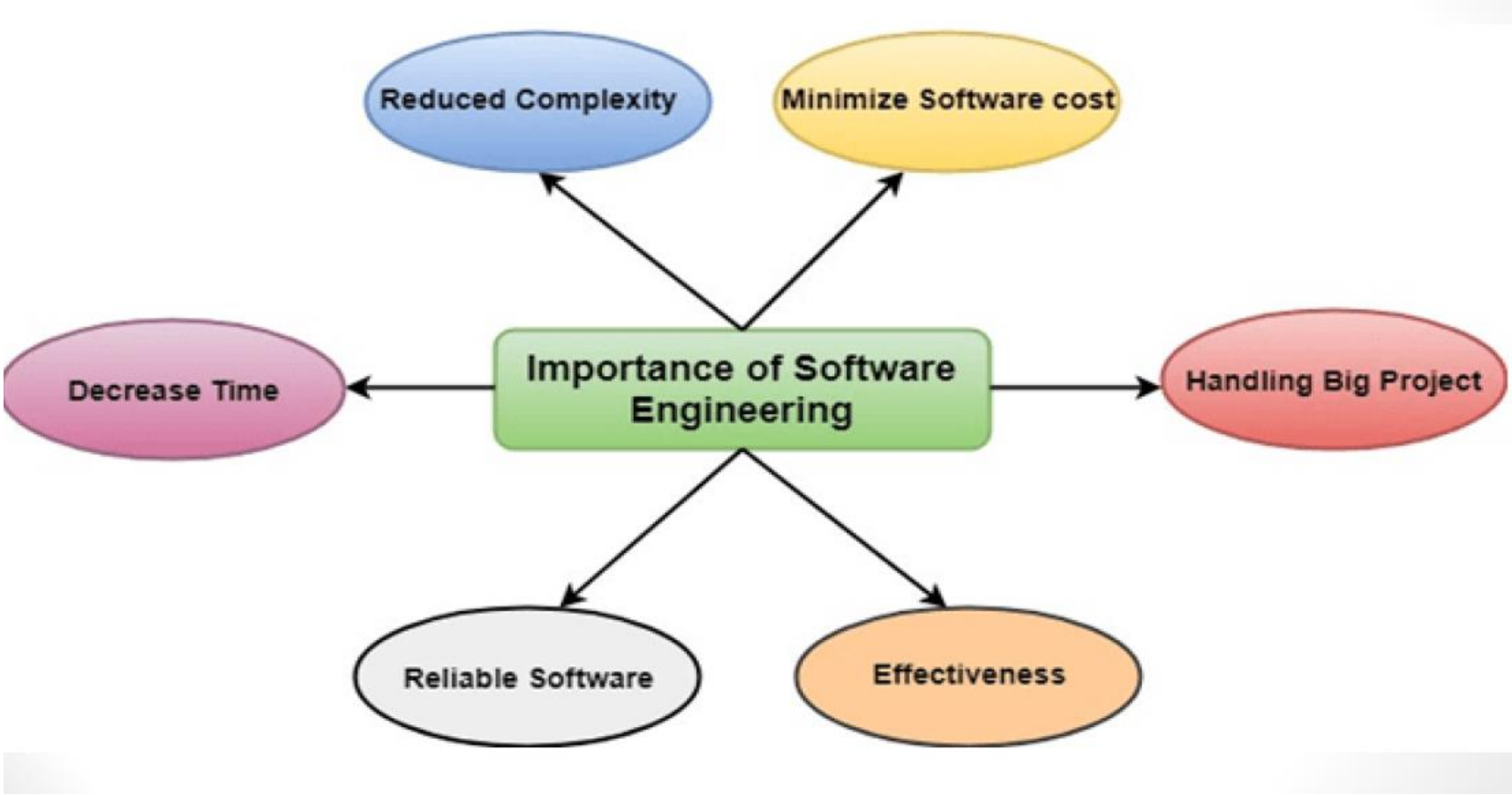
- The term **software engineering** is the product of two words, **software**, and **engineering**.
- The **software** is a collection of integrated programs.
- Software subsists of carefully-organized instructions and code written by developers on any of various particular computer languages.
- Computer programs and related documentation such as requirements, design models and user manuals.
- **Engineering** is the application of **scientific** and **practical** knowledge to **invent, design, build, maintain**, and **improve frameworks, processes, etc**.

What is Software Engineering?

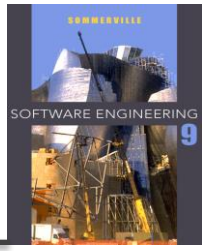


- **Software Engineering** is an engineering branch related to the evolution of software product using systematic well-defined scientific principles, techniques, and procedures.
- The result of software engineering is an effective and reliable software product.

✧ Importance of Software Engineering



✧ Importance of Software Engineering



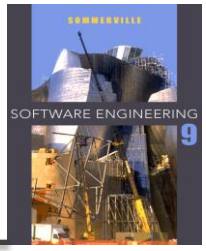
✧ **(1) Reduces complexity:**

✧ Software engineering divides big problems into various small issues. And then start solving each small issue one by one.

✧ All these small problems are solved independently to each other and then integrated together to produce the software product.

Big Problem → Small Problems → Solve Each Problem → Integrate → Complete Software

✧ Importance of Software Engineering

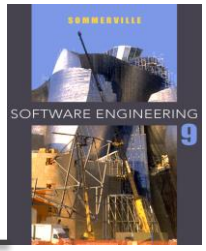


✧ **(2) To minimize software cost:**

✧ A lot of resources are required to develop large-scale software systems, such as: manpower, software licenses, hardware…etc.

✧ With Software Engineering, the project is planned and estimated before and during development, helping the company **control costs and avoid unnecessary expenses**.

✧ Importance of Software Engineering



✧ **(3) To decrease time:**

✧ One of the main criteria to measuring project success and yet the most challenging is delivering software projects on time.

✧ Schedule issues are the main reason for conflicts on projects, especially during the second half of projects where the actual implementation takes place to produce the final working product.

✧ Software Engineering helps decrease development time by providing systematic processes for planning, scheduling, coordinating, and monitoring project activities, ensuring that the software is completed and delivered on time.